Explanation Stability under Class-Incremental Network Intrusion Detection: An Oracle-Free Drift Protocol and a Trust-Governance Mechanism

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Abstract—A deployed network intrusion detection system (NIDS) must keep learning new attack families, and analysts increasingly rely on per-decision feature attributions to triage, tune and audit its alerts. We study a property that sits between these two demands: whether the explanations of an incrementally-trained NIDS stay trustworthy as new attacks are added. We find that accuracy-preserving models can keep classifying an old attack correctly while the features they rely on to do so silently change, and that for some attack families the model never had a faithful, concentrated rationale in the first place. We contribute three things. First, an oracle-free silent-drift protocol that measures step-to-step attribution change without an all-data reference model, together with a forgetting-versus-data-shift decomposition and a faithfulness check. Second, OFRA (Oracle-Free Routed Adapters), a detector that learns each attack family as a small frozen low-rank adapter routed at inference without a task oracle; OFRA is a Pareto operating point that trades 8–16 pp accuracy for near-zero forgetting ($\sim 3\times$ lower than the strongest baseline, geometric mean over five datasets), an order of magnitude fewer per-task parameters, and 0% attribution drift on the isolated path by construction. We show this 0% is only partial: after the oracle-free router the shipped decision still drifts 2.3% pooled. Third, ETG (Explanation-Trust Gate), a per-family trust state machine driven by a deletion-faithful rationale-mass test that governs whether each family's explanation may be trusted, routed to a human, or declared unexplainable—without ever changing the model weights. Across five IDS datasets with five seeds and Holm-corrected significance testing, ETG certifies 34 of 39 families, refuses five ill-posed ones, and suppresses two silent-drift alarms that a plain drift monitor would raise on families with no faithful rationale. An ablation shows the deletion-faithful engine is load-bearing: 19 of 39 trust decisions flip under ordinary gradient attribution. We report all results honestly: OFRA is a different operating point, not an overall winner, and ETG is at proof-of-concept maturity. A later, separate prediction–retention study reports developmental final-BAcc gains for the separate gate over the same-data single classifier of 17.0 pp on UNSW-NB15 and 24.8 pp on CIC-IDS-2017 at a 1% calibration target (five seeds each, both $p < 0.01$, as reported by the lead), but also yields a trade-off rather than an accuracy winner: in the lead-reported CSE18 seeds 5–9 summary, AR lowers signed forgetting but costs about 4–5 pp mean incremental balanced accuracy relative to exposure-matched Replay-BS. Final BAcc superiority is not confirmed at either 2% or 1% calibration targets (the latter advantage is 1.9 pp, not significant). This same-dataset follow-up neither establishes equivalence nor validates explanation governance; its initially withheld seeds are not an independent dataset replication.

Index Terms—intrusion detection, class-incremental learning, explainable AI, explanation stability, feature attribution, low-rank adaptation, trust governance.

I. Introduction

Network intrusion detection systems operate in a non-stationary world. New attack families appear continuously, and retraining a detector from scratch on every arrival is costly and often impossible because old traffic is unavailable for privacy or storage reasons. The practical setting is therefore class-incremental learning (Class-IL) [1]: the model is updated task by task, at inference there is no oracle telling it which task a sample belongs to, and any memory of old data is tightly bounded. The dominant failure mode studied in this setting is catastrophic forgetting [2]: accuracy on earlier attacks collapses as new ones are learned.

Almost all continual-learning work, in intrusion detection and elsewhere, optimises the same two numbers—accuracy and forgetting—and asks a single question of every update: did the prediction survive? We argue that in a security setting this question is incomplete. Analysts do not consume raw predictions; they consume explanations. A feature attribution tells an operator why an alert fired, drives rule tuning, and is retained for forensic and compliance review. If the model keeps predicting an old attack correctly but the features it now relies on to do so have quietly changed, the explanation an analyst trusted yesterday no longer describes today's model—and nothing in an accuracy dashboard reveals it. We call this silent explanation drift. That attributions can be fragile under small perturbations is known [3]; what has not been studied is whether they survive the incremental updates of a deployed detector.

This paper takes explanation stability as a first-class, measurable property of Class-IL intrusion detection and makes three contributions.

1) An oracle-free silent-drift protocol (Section V). A deployed detector never has an all-data "oracle" model to compare against, so we measure drift between successive increments: for each preserved old class we compare its top-k attribution set at update t against $t-1$ by Jaccard overlap, cosine and Kendall-τ, and flag silent drift when accuracy is preserved but the top-15 attribution Jaccard falls below 0.70. We pair this with a forgetting-versus-data-shift decomposition and a deletion-based faithfulness check so that a drop in overlap cannot be dismissed as "the data simply changed."
2) OFRA (Oracle-Free Routed Adapters, Section IV), a detector whose isolated paths are stable by construction. A frozen shared encoder feeds a per-attack-family rank-8 adapter that is frozen once learned, selected at inference by an oracle-free router. OFRA is deliberately reported as a Pareto operating point, not a winner: it pays 8–16 pp accuracy in exchange for near-zero forgetting, roughly 10× fewer parameters per task, and 0% attribution drift on the isolated path. We then show this headline 0% is partial—after the oracle-free router the shipped decision still drifts 2.3% pooled—and identify that decision-level residue as the blind spot our third contribution governs.
3) ETG (Explanation-Trust Gate, Section VI), a per-family trust state machine that governs the consumption of explanations over the incremental stream and never touches the model weights. Its engine is a deletion-faithful rationale-mass test that first asks whether a faithful, concentrated rationale exists at all; families that pass are certified and monitored for drift, families that fail are honestly declared unexplainable, and drifted families are routed to a human rather than silently repaired.

a) Later bounded RQ2 contribution.: The supplemental benign-stream study compares gate/family retention choices, including the failed fixed-KD+WA assessment and the subsequent AR round. Its contribution is an evidence-bounded retention–plasticity comparison, not a claim that KD, DER-style replay, weight alignment or an extra classifier is novel. The lead-reported final AR results fail accuracy superiority over Replay-BS and preserve the historical negative outcomes. These experiments do not close the locked explanation-drift, governance or substantive-innovation requirements; unreported variants remain incomplete rather than being omitted as unsuccessful candidates.

We evaluate on five public IDS datasets—NSL-KDD [4], UNSW-NB15 [5], CIC-IDS-2017 [6], CIC-IDS-2018 [7] and NF-ToN-IoT-v2 [8]—with five seeds and Holm-corrected paired tests. Throughout, we state the honest scope of every claim: where a number is a structural property we say so, where it is not significant we report it as such, and where an interpretation did not survive a control we decline to make it.

II. Related Work

A. Class-incremental learning

Continual-learning methods fall into three broad families [1]: regularization approaches that penalise change to important weights (EWC [9], LwF [10]); replay approaches that rehearse stored or generated exemplars (iCaRL [11], GDumb [12], reservoir and gradient-based selection [13]); and parameter-isolation approaches that allocate separate capacity per task. Recent work adapts large frozen backbones with lightweight prompts or low-rank modules (L2P [14], CODA-Prompt [15], CLLoRA [16]), building on LoRA [17]. Our detector belongs to the parameter-isolation family but is designed for a tabular, centralized IDS stream rather than vision.

B. Relation to ECCV-2024 PILoRA

A method named PILoRA [18] was published concurrently in a federated, vision setting. Because the acronym collides with an earlier working name for our detector, we clarify the relationship explicitly and have adopted the distinct name OFRA. The two designs are opposite on their central axis. PILoRA maintains a single unified incremental LoRA and classifies by nearest prototype; its authors state that their method "does not require training new LoRA modules at different tasks and selecting based on similarity at inference," and position that per-module-selection strategy as the approach they avoid. OFRA does exactly the avoided thing on purpose: it keeps separate frozen per-family adapters and selects one at inference with an oracle-free router, in a centralized tabular-IDS setting, for the purpose of explanation stability rather than accuracy under federated heterogeneity. The overlap is the generic "LoRA-for-Class-IL" umbrella, which is now common to many methods; the specific mechanism is not shared.

C. Continual learning for intrusion detection

Continual NIDS work [19]–[21] has concentrated on preserving detection accuracy under streaming attacks. Streaming evaluation is comparatively rare, and to our knowledge no prior NIDS study asks whether the model's explanation of an old attack survives incremental updates.

D. Explanation stability and faithfulness

Feature-attribution methods such as SHAP [22] are widely used to interpret detectors, and their fragility to input perturbation is documented [3]. Faithfulness—whether an attribution actually reflects the features the model uses—is typically probed by deletion or erasure tests [23], [24]. We reuse this deletion idea not as a

saliency-quality metric on a static model, but as the gating engine of a trust state machine that runs over an incremental stream. Work on explanation continuity that assumes a known task identifier at test time is the closest neighbour; our setting is oracle-free, which is the realistic deployment case.

III. Problem Formulation

We observe a sequence of tasks $\mathcal{T}_1, \ldots, \mathcal{T}_T$, where task $\mathcal{T}_t$ introduces one or more new attack families. After training on $\mathcal{T}_t$ the model must classify all families seen so far, with no task identifier at inference (Class-IL) and a per-family exemplar budget of 50. For a preserved old family c (accuracy drop $\Delta_c > -5\%$ from its value when first learned) let $A_t(c)$ be the set of its top-k attributed features at increment t. We define silent drift for c at t as

$$\text{Acc preserved} \ \wedge \ J\big(A_t(c), A_{t-1}(c)\big) < \tau, \qquad (1)$$

with top-15 features and Jaccard threshold $\tau = 0.70$; we report robustness to both choices. Crucially, A_t and A_{t-1} are two deployed models one increment apart—no all-data oracle is required, which matches a real NIDS that never has such a model.

IV. OFRA: The Detector Foundation

OFRA (Fig. 1) has four parts.

Frozen shared encoder. A two-layer MLP ($d \to 128 \to 128$) is trained on the first task and then frozen. Because the representation never moves, an old family's feature space—and therefore its attribution—cannot drift as a side effect of later encoder updates.

Per-family adapters. Each new attack family is learned as its own rank-8 LoRA adapter [17] on top of the frozen encoder (roughly 1.2K trainable parameters), and frozen once learned. Learning a new attack therefore cannot overwrite an old one along its isolated path.

Oracle-free router. At inference a DP-Means [25] router selects the family adapter by prototype distance, without a task identifier—the realistic deployment case.

Bounded exemplar memory. Fifty raw exemplars per family support replay and calibration.

The design gives a structural property: on the isolated path, an old family's frozen adapter cannot change, so its attribution is invariant by construction. We stress that this is a property of the isolated path, not a guarantee about the deployed decision (Section VIII-A), and we do not claim OFRA is the most accurate detector—Section VIII-B shows it is not.

V. Oracle-Free Silent-Drift Protocol

For every preserved old family at every increment we compute attributions with SHAP and an input-gradient estimator, rank features, and compare the top-15 sets across the two most recent increments by Jaccard, cosine and Kendall-τ. Two safeguards keep the signal honest. A forgetting-versus-data-shift decomposition separates attribution change caused by the model from change caused by input distribution shift, so a low overlap cannot be waved away as a data artefact. A deletion-based faithfulness check confirms that the ranked features are the ones the model actually uses. The protocol needs only two consecutive deployed checkpoints; it never constructs the all-data reference model that continuity methods assume.

VI. ETG: Explanation-Trust Gate

OFRA's structural stability covers the isolated path, but not the decision the router ships (Section VIII-A). Rather than try to prevent that residual drift—which we show is hard to do without paying accuracy (Section VIII-B)—ETG governs it. ETG changes no weights; it governs how much each family's explanation may be trusted.

A. Engine: rationale mass

Before drift can be tracked, a faithful rationale must exist. For family c we rank features by occlusion importance (the drop in the class-c softmax probability when a feature is zeroed) and define its rationale mass as the joint deletion-gap of the top-15 occlusion features. A family is admitted only if its mass exceeds a random-15 null (the 95th percentile of masses from random feature sets). This metric is reproducible, stable under a split-half check, near-zero for an anti-rationale (bottom-15) baseline, and random-controlled. Where mass is essentially zero, explanation stability is ill-posed and a drift alarm there would be noise.

B. State machine

Each family carries a persistent state (Fig. 2):

$$\text{UNCERTIFIED} \to \{\text{CERTIFIED-STABLE}, \text{UNEXPLAINABLE}\} \to \text{DRIFTED} \to \{\text{re-certify, UNEXPLAINABLE}\}.$$

On admission, a family whose mass clears the null becomes CERTIFIED-STABLE and its reference rationale is frozen; otherwise it is UNEXPLAINABLE. At a later increment, if accuracy is preserved but the top-15 Jaccard against the certified reference falls below 0.70, the family becomes DRIFTED. A drifted family whose mass later recovers may return to CERTIFIED-STABLE; otherwise it is demoted.

C. Action policy

The states drive operator-facing actions. A CERTIFIED-STABLE explanation is audit-grade and eligible for auto-disposition. A DRIFTED alert is force-routed to a human and its stale rationale is withheld. An UNEXPLAINABLE family still raises its detection alert—detection is untouched—but its unreliable explanation is suppressed and no drift alarm is emitted for it. The auto-maintained per-family trust ledger (family, increment, mass, action) is the artefact a SOC actually consumes.

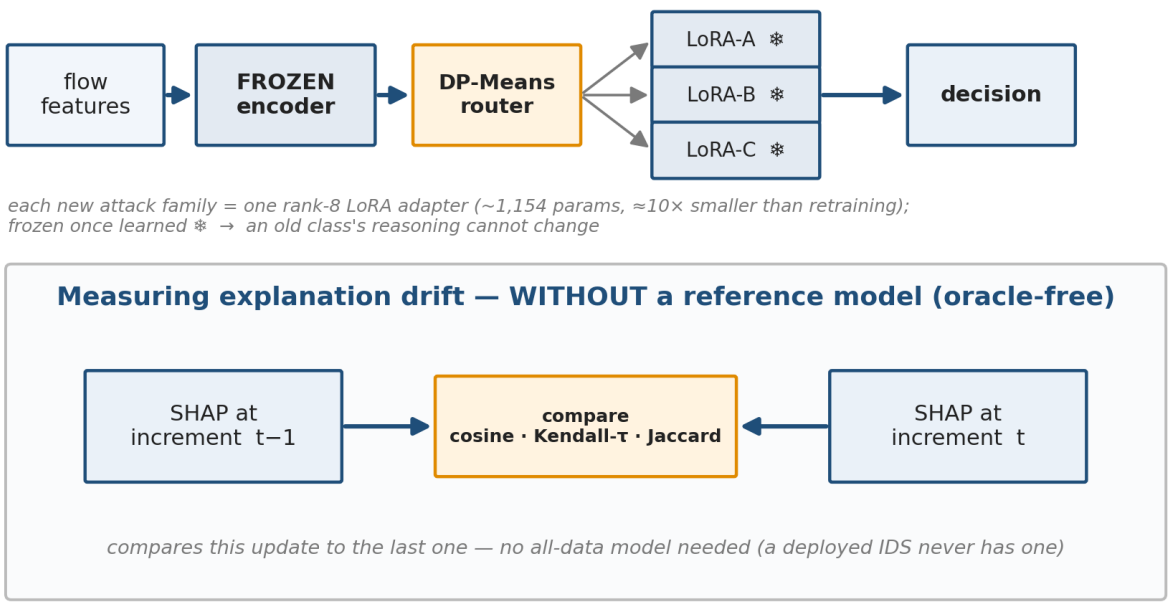


Fig. 1. OFRA. A frozen shared encoder feeds a per-attack-family rank-8 adapter, frozen once learned and selected at inference by an oracle-free DP-Means router. Drift is measured by comparing attributions at increment t against $t-1$, with no all-data reference model.

VII. Experimental Setup

We use five datasets—NSL-KDD, UNSW-NB15, CIC-IDS-2017, CIC-IDS-2018 and NF-ToN-IoT-v2—in attack-family-incremental order, each with five seeds. We compare eight methods spanning the three CL families: a naive lower bound, EWC [9], LwF [10], Replay, DP-Means Replay, iCaRL [11], a recent frozen-backbone baseline (NCM-Frozen), and OFRA. Forgetting is the mean drop of each old family's accuracy from its first-learned value; significance uses paired tests with Holm correction over the family of comparisons [26], [27].

VIII. Results

A. Silent drift: the 0% is isolated-path only

Accuracy-preserving shared-parameter baselines silently drift 3–100% depending on the dataset, with every baseline separated from OFRA at $p < 0.001$ and Cohen's $d \approx 6$. On the isolated path OFRA drifts 0% by construction. This headline is real but partial: the deployed decision is produced after the oracle-free router, which can re-route old-attack inputs as new families are added. Measuring drift at the decision level (Table I, Fig. 3) gives 2.3% pooled over 26 runs—6.7% on NSL-KDD and 6.2% on CIC-IDS-2017, 0% on UNSW-NB15 and CIC-IDS-2018. This residual is exactly the blind spot ETG governs.

TABLE I
Silent-drift rate (%). Baselines are shared-parameter models; the OFRA column is the honest deployed (decision-level) drift. OFRA's isolated path is 0% by construction.

Dataset	Naive	Replay	DP-Means	OFRA
NSL-KDD	53	15	28	6.7
UNSW-NB15	61	37	43	0.0
CIC-IDS-2017	79	82	85	6.2
CIC-IDS-2018	100	81	92	0.0
NF-ToN-IoT-v2	17	3	6	1.2
OFRA pooled (decision-level)				2.3

B. Forgetting and the accuracy trade-off

OFRA lowers forgetting by a geometric-mean factor of 3.1× against the strongest baseline (Table II), significant on four of five datasets after Holm correction. We report CIC-IDS-2018's 1.3× as not significant ($p = 0.10$) rather than hiding it. This comes at an honest accuracy cost of 8–16 pp and with roughly an order of magnitude fewer trainable parameters per family. We do not claim state-of-the-art accuracy: OFRA is a different operating point on the accuracy–stability plane, not an overall winner.

We also asked whether a new training-time mechanism could simply prevent the decision-level drift. Across our five datasets and an eight-angle search over three custom regularisers, frozen isolation empirically Pareto-dominated every prevention configuration we tried—it reached 0% isolated-path drift at no worse accuracy and lower forgetting than any constrained alternative. We treat this as an empirical observation on our benchmark, not a proof of impossibility; it is why we chose to govern the residual

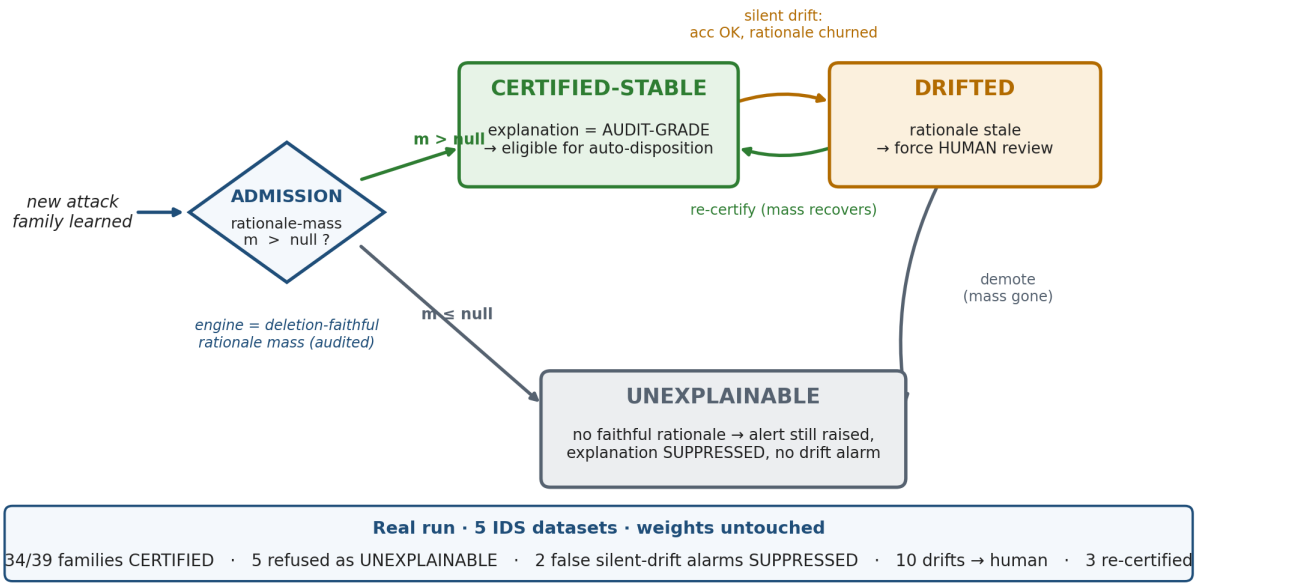


Fig. 2. ETG state machine. A family is admitted only if a faithful, concentrated rationale exists (rationale-mass above a random null); certified families are monitored for drift; ill-posed families are declared unexplainable; drifted families are routed to a human. ETG never changes model weights.

TABLE II
Forgetting reduction of OFRA versus the strongest baseline (five seeds, Holm-corrected), and OFRA's accuracy cost.

Dataset	Forgetting reduction	p_{Holm}	Acc. cost
NSL-KDD	6.6×	0.009	−16.0 pp
UNSW-NB15	3.5×	8.4×10^{-6}	−7.7 pp
CIC-IDS-2017	2.9×	0.027	−10.6 pp
CIC-IDS-2018	1.3×	0.10 (n.s.)	−7.8 pp
NF-ToN-IoT-v2	2.3×	—	—
Geometric mean	**3.1**×		

TABLE III
ETG per-family trust ledger (five datasets, five seeds). ETG never changes model weights.

Dataset	Certified	Refused	Alarms supp.	Drifts→human
NSL-KDD	5/5	0	0	2
UNSW-NB15	9/10	1	1	1
CIC-IDS-2017	6/7	1	0	2
CIC-IDS-2018	7/7	0	0	4
NF-ToN-IoT-v2	7/10	3	1	1
Total	34/39	5	2	10

drift rather than fight it.

C. ETG: the trust ledger

Table III is the artefact ETG produces. Across the five datasets it certifies 34 of 39 families, refuses five as unexplainable, suppresses two silent-drift alarms on families with no faithful rationale, and routes ten real drifts to a human (with three later re-certified). The refusals and suppressions are work a plain accuracy monitor or drift monitor cannot do: they are the cases where an alarm would be noise, or where a certificate would be dishonest.

The engine is load-bearing. Replacing the deletion-faithful engine with ordinary input-gradient attribution changes 19 of 39 admission decisions (occlusion admits 34/39; input-gradient admits 19/39). The faithfulness of the attribution is not an implementation detail—it decides half the ledger.

Stability across seeds. The admission partition is stable: NSL-KDD, CIC-IDS-2018 and NF-ToN-IoT-v2 have zero standard deviation across five seeds, while UNSW-NB15 (9.2 ± 0.4) and CIC-IDS-2017 (6.2 ± 0.8) wobble only on families whose mass sits near the null.

Why govern, not auto-repair. A natural alternative is to penalise the model to restore a drifted family's rationale. We implemented a triggered, family-local, occlusion-anchored repair. Globally it looks almost free—accuracy barely moves—but a per-family audit shows it silently destroys a rare attack family: repairing UNSW-NB15 Fuzzers/Reconnaissance drops Shellcode accuracy by $\sim$ 45 pp, and CIC-IDS-2017 repairs cost Infiltration $\sim$14 pp. Auto-repair is therefore unsafe on this data; ETG routes drift to a human instead.

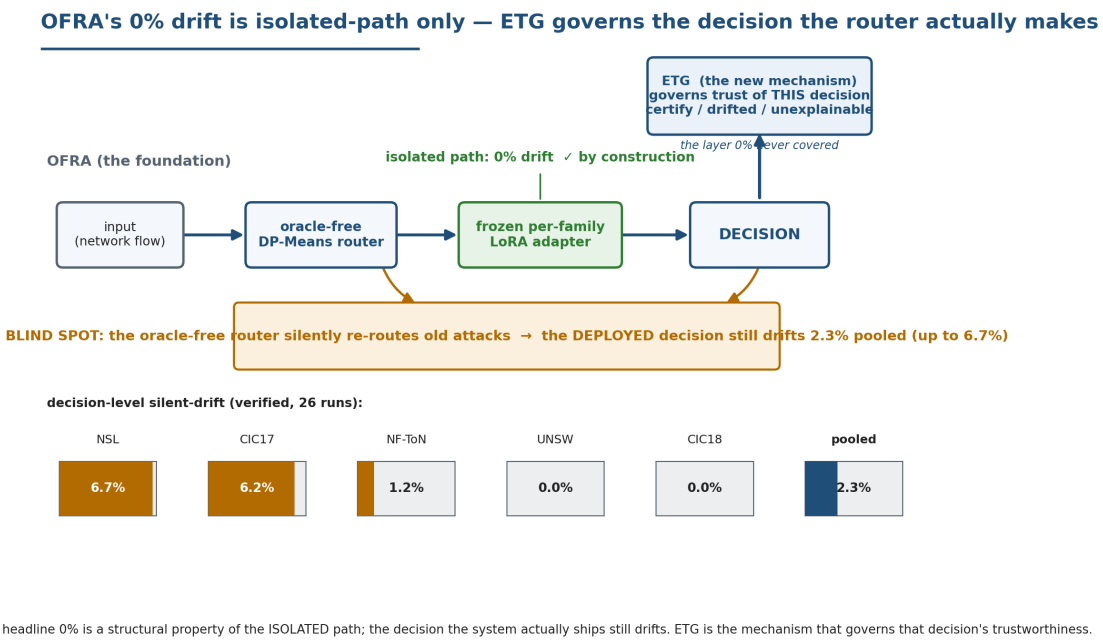


Fig. 3. OFRA's 0% is a structural property of the isolated path; after the oracle-free router the shipped decision still drifts 2.3% pooled (up to 6.7%). ETG governs the trustworthiness of that decision.

IX. Discussion

The positioning of this work is narrow on purpose. OFRA is oracle-free, unlike explanation-continuity methods that assume a known task identifier. ETG carries per-family persistent state over the incremental stream, unlike stateless per-alert trust triage on a static model, and it is gated by a faithfulness test so that it declares "unexplainable" rather than certifying an ill-posed family. It is a governance layer on top of the detector, not a drift-prevention algorithm, and it targets exactly the decision-level residue that frozen isolation leaves uncovered.

X. Supplemental prediction–retention trade-off

This later, bounded RQ2 study compares a refreshed benign–attack gate with an attack-family learner. AR removes gate distillation while retaining DER-style family replay and weight alignment; it is distinct from frozen OFRA isolation and from ETG. Tables IV–V reproduce the lead's rounded CSE18 follow-up computation. AR is not confirmed as a final-accuracy improvement over Replay-BS: lower forgetting costs about 4–5 percentage points of mean incremental BAcc. Non-significance is not equivalence. These new-seed results do not replace the historical OFRA tests, establish an explanation-drift or governance benefit, or close the locked RQ1/RQ3 and substantive-innovation requirements. Full remaining-method reporting remains pending.

XI. Limitations

We state the boundaries plainly. ETG governs explanation consumption; it does not improve detection accuracy, and its value assumes explanations are actually used and must be trustworthy. Its cost is not zero: drifted and unexplainable families are routed to humans, a coverage/analyst-load cost that should be measured over a live stream. Rationale mass is computed on class-mean probes; per-sample governance is future work. The evidence for ETG is at proof-of-concept maturity—a small number of alarms per dataset—and we deliberately do not claim that a drift alarm localises which feature the model migrated to, because that reliance-migration interpretation did not survive a random-controlled test; a DRIFTED verdict is only a conservative "stop auto-trusting" signal. Finally, a sixth dataset (CIC-IoT-2023) is not yet included.

XII. Conclusion

We treated explanation stability as a measurable property of class-incremental intrusion detection. An oracle-free protocol makes silent explanation drift visible without an all-data reference model; OFRA is an oracle-free detector whose isolated paths are stable by construction, offered as a Pareto operating point rather than an accuracy winner; and ETG governs the trustworthiness of the deployed decision that OFRA's structural 0% does not cover, certifying families whose rationale is faithful, refusing those whose rationale is ill-posed, and routing drift to a human rather than silently retraining. Future work will formalise the certificate and its thresholds, lift certification from the isolated path to the routed decision, and evaluate alarm latency and false-alarm rate on a streaming deployment.

TABLE IV
CSE18, paired seeds 5–9 ($n = 5$): retention–plasticity trade-off. Blind seeds 5–9; externals evaluated from saved checkpoints, no retraining (1% panel).

Target	Method	Mean inc. BAcc	Final BAcc	Forget (pp)	ADR	Test FPR
2%	AR	81.7	77.4	−0.2	95.0	1.97
	AA-CIL	84.7	77.4	4.1	95.0	1.97
	Replay-BS	86.2	77.5	8.3	94.3	1.99
	Freq	69.9	61.5	14.2	75.8	2.00
	WACB	65.5	50.1	19.9	50.7	2.03
1%	AR	80.9	76.3	−0.8	94.5	1.00
	AA-CIL	83.7	76.2	3.8	94.5	1.00
	Replay-BS	84.7	74.4	8.2	93.3	1.01
	Freq	63.9	58.3	11.4	68.2	1.00
	WACB	50.2	28.7	15.8	18.8	1.03

Values are percentages except signed forgetting in percentage points; negative forgetting is retained, not clipped. Rounded means are transcribed from the lead's recomputation recorded in work-order Sections 71–72, except the completed 2% ADR and test-FPR columns, which are arithmetic means of the five saved final values per method in the published Section 64 seed releases. No confusion matrices or statistical tests were recomputed for this completion. The 2% panel derives from the 25 published decision cells; the 1% panel combines saved AR/AA/BS operating points with Freq/WACB checkpoint evaluation. No model was retrained for the 1% panel. "Blind seeds" describes the initially withheld seed results, not an independent dataset: the design followed unsealing of the earlier CSE round, uses the same exposed dataset, and complete single-seed interim releases preceded this final summary. No early stopping, tuning, or criterion changes followed those releases.

TABLE V
CSE18 lead-reported paired tests at 1% ($n = 5$). Differences are AR minus reference, in percentage points.

Reference	Endpoint (alternative)	Difference	One-sided t p	Exact sign-flip p
Freq	Final BAcc (>)	18.0	.045	.031
Freq	Mean inc. BAcc (>)	17.0	.005	.031
WACB	Final BAcc (>)	47.6	.001	.031
WACB	Mean inc. BAcc (>)	30.7	.003	.031
Replay-BS	Final BAcc (>)	1.9	.059	.062

Rounded values from work-order Sections 71–72; paired t has four degrees of freedom, and sign-flip enumerates all 32 sign assignments. These are reported tests, not new tests executed during typesetting. At 1%, the lead reports AR forgetting differences of −12.1pp versus Freq and −16.5pp versus WACB; harm is not significant against the lowest-forgetting external (Freq). Thus the external half of H-low passes for AR, but the Replay-BS superiority half fails, so the conjunction is not confirmed. This is not an equivalence finding. At 2%, the registered final-BAcc superiority criterion also fails (AR minus BS -0.08 ± 3.11pp; reported one-sided $p \geq .52$), while the other three criteria pass. No new round or selection is implied. AR has the lowest mean forgetting among these five available decision methods, not among unreported variants.

The later AR follow-up reinforces the need to report operating points rather than an overall winner. The lead's CSE18 summary places final BAcc at 77.4% for AR and 77.5% for Replay-BS at the 2% target, with lower AR forgetting but about 4–5 pp lower mean incremental BAcc. At 1%, AR exceeds Freq and WACB, yet the 1.9 pp advantage over Replay-BS is not significant, so H-low remains unconfirmed. No equivalence, causal gate explanation, or new method-selection success follows. These tests are distinct from the historical OFRA CIC-IDS-2018 result ($p = 0.10$, not significant), and neither changes it. The published follow-up tables identify the lead's rounded recomputation; remaining-method descriptive reporting is still pending. The study's predictive trade-off does not by itself complete RQ1/RQ3 or the substantive-innovation gate.

Code, results, and the source of this paper are available at https://github.com/1838904818/Class-IL.

References

[1] M. Masana, X. Liu, B. Twardowski, M. Menta, A. D. Bagdanov, and J. van de Weijer, "Class-incremental learning: survey and performance evaluation on image classification," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 45, no. 5, pp. 5513–5533, 2023.

[2] R. M. French, "Catastrophic forgetting in connectionist networks," Trends in cognitive sciences, vol. 3, no. 4, pp. 128–135, 1999.

[3] A. Ghorbani, A. Abid, and J. Zou, "Interpretation of neural networks is fragile," in Proceedings of the AAAI Conference on Artificial Intelligence, 2019.

[4] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the kdd cup 99 data set," in 2009 IEEE symposium on computational intelligence for security and defense applications. IEEE, 2009, pp. 1–6.

[5] N. Moustafa and J. Slay, "Unsw-nb15: a comprehensive data set for network intrusion detection systems (unsw-nb15 network data set)," in 2015 military communications and information systems conference (MilCIS). IEEE, 2015, pp. 1–6.

[6] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in ICISSp, 2018, pp. 108–116.

[7] I. Sharafaldin, A. H. Lashkari, S. Hakak, and A. A. Ghorbani, "Developing realistic distributed denial of service (ddos) attack dataset and taxonomy," in 2019 International Carnahan Conference on Security Technology (ICCST). IEEE, 2019, pp. 1–8.
[8] M. Sarhan, S. Layeghy, N. Moustafa, and M. Portmann, "NetFlow datasets for machine learning-based network intrusion detection systems," Big Data Technologies and Applications (EAI), 2021.
[9] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska et al., "Overcoming catastrophic forgetting in neural networks," Proceedings of the national academy of sciences, vol. 114, no. 13, pp. 3521–3526, 2017.
[10] Z. Li and D. Hoiem, "Learning without forgetting," IEEE transactions on pattern analysis and machine intelligence, vol. 40, no. 12, pp. 2935–2947, 2017.
[11] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "icarl: Incremental classifier and representation learning," in Proceedings of the IEEE conference on Computer Vision and Pattern Recognition, 2017, pp. 2001–2010.
[12] A. Prabhu, P. H. S. Torr, and P. K. Dokania, "GDumb: A simple approach that questions our progress in continual learning," in European Conference on Computer Vision. Springer, 2020, pp. 524–540.
[13] R. Aljundi, M. Lin, B. Goujaud, and Y. Bengio, "Gradient based sample selection for online continual learning," in Advances in Neural Information Processing Systems, vol. 32, 2019.
[14] Z. Wang, Z. Zhang, C.-Y. Lee, H. Zhang, R. Sun, X. Ren, G. Su, V. Perot, J. Dy, and T. Pfister, "Learning to prompt for continual learning," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 139–149.
[15] J. S. Smith, L. Karlinsky, V. Gutta, P. Cascante-Bonilla, D. Kim, A. Arbelle, R. Panda, R. Feris, and Z. Kira, "CODA-Prompt: COntinual decomposed attention-based prompting for rehearsal-free continual learning," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.
[16] J. Liang et al., "CL-LoRA: Continual low-rank adaptation for rehearsal-free class-incremental learning," arXiv preprint arXiv:2505.24816, 2025.
[17] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in International Conference on Learning Representations (ICLR), 2022.
[18] H. Guo, F. Zhu, W. Liu, X.-Y. Zhang, and C.-L. Liu, "PILoRA: Prototype guided incremental LoRA for federated class-incremental learning," in European Conference on Computer Vision (ECCV), 2024.
[19] S. K. Amalapuram, B. R. Tamma, and S. S. Channappayya, "Augmented memory replay-based continual learning approaches for network intrusion detection," in IEEE INFOCOM 2022-IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS). IEEE, 2022, pp. 1–6.
[20] X. Zhang, R. Zhao, Z. Jiang, H. Chen, Y. Ding, E. C. H. Ngai, and S.-H. Yang, "Continual learning with strategic selection and forgetting for network intrusion detection," arXiv preprint arXiv:2412.16264, 2024.
[21] M. S. Islam, L. G. Jaimes, and A. S. Dina, "MA-IDS: Multi-agent RAG framework for IoT network intrusion detection with an experience library," arXiv preprint arXiv:2604.05458, 2026.
[22] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems (NeurIPS), 2017.
[23] V. Petsiuk, A. Das, and K. Saenko, "RISE: Randomized input sampling for explanation of black-box models," in British Machine Vision Conference (BMVC), 2018.
[24] J. DeYoung, S. Jain, N. F. Rajani, E. Lehman, C. Xiong, R. Socher, and B. C. Wallace, "ERASER: A benchmark to evaluate rationalized NLP models," in Proceedings of the Association for Computational Linguistics (ACL), 2020.
[25] B. Kulis and M. I. Jordan, "Revisiting k-means: New algorithms via bayesian nonparametrics," arXiv preprint arXiv:1111.0352, 2012.
[26] J. Cohen, "Statistical power analysis for the behavioral sciences." Routledge, 1988.
[27] C. Wohlin, P. Runeson, M. Höst, M. C. Ohlsson, B. Regnell, and A. Wesslén, Experimentation in Software Engineering. Springer Berlin Heidelberg, 2012.